

Closed on Schedule, Not Unloaded

An Audit of the Loading Dock's Roller-Door Sensor Timer Against Drivers' Own Paper Dockets

Verity Marris · Casimir Beng — School of Emergent Priorities

The problem

Facilities retired the loading dock's handwritten arrival log this term, replacing it with a roller-door sensor that times every delivery from door-up to door-down and feeds a live "median dock turnaround" figure to the Estates Efficiency Dashboard. Whether that interval tracks unloading, or just the door, is untested.

Research questions

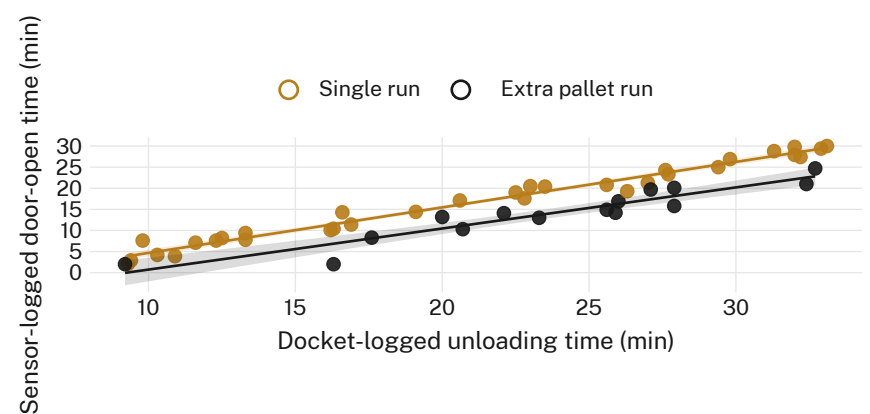
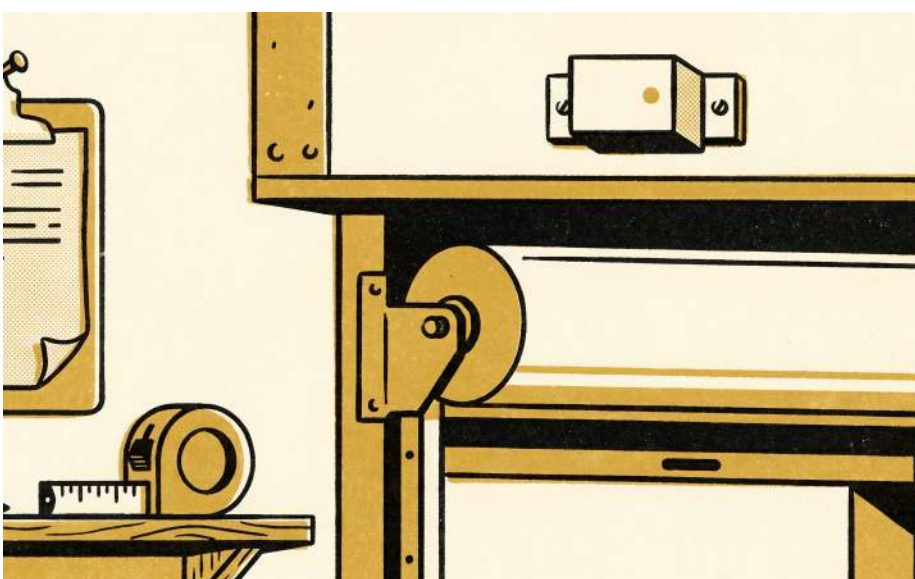
- Does sensor-logged door-open duration track drivers' independently kept dockets?
- Did the reported median turnaround shift at the week of the switch, or only afterward?
- Does any gap widen for deliveries requiring a second pallet run?

Study design

- n = 48 deliveries · 8 dock weeks spanning the sensor's install · paired sensor and docket times
- linear fit, sensor duration on docket duration, by run type
- extra-pallet-run flag and tolerance band pre-registered with the dock roster

Findings

Sensor and docket times move together ($r = 0.93$), but the sensor consistently reports less. The reported median turnaround dropped the week the sensor went live, with no matching change in the docket record.



Across 48 logged deliveries in Weeks 3-10, sensor-logged door-open time undercut drivers' docket-logged unloading time by a median of 5.5 minutes, widening to a median of 9.3 minutes ($n = 15$) when a delivery required a second pallet run.

Implications

The dashboard's turnaround figure now times the door, not the delivery. Crew report closing the door once the first pallet clears, well before the last carton is off the ramp — a habit the switch rewarded without asking for it. Next: pair the sensor with a last-carton scan before retiring the docket entirely.

Selected references

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Delivery records de-identified by supplier account only; driver rosters were not retained. We thank the loading dock team for tolerating an eight-week paper trail.

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